

Nikki Phach

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Skills

TypeScript, Python, React, XState, SQL, Flask, Phaser, TailwindCSS, @shadcn/ui, Vitest, Zod, NumPy, Flask, Scikit-learn, Pandas, Responsive Web Design, Docker, npm, HTML5, CSS, C, PyTorch, Machine Learning, Computer Vision, Natural Language Processing, Data Structures/ Algorithms, Unit Testing, Web Scraping, JavaScript, Data Processing/ Visualization, Git/ GitHub, Database Design, APIs, Racket, LaTeX, Java, Linux/ Unix

Work Experience

Konami Gaming Inc.

April 2025 - present

Software Engineer I

- Enabled promotional free spin capability for 60+ online casinos by developing a universal feature that integrates with a third-party content provider's API and UI wrapper.
- Ensured compliance with Brazil gaming regulations by introducing a user confirmation flow for automatic spins within gameplay.
- Improved flexibility of RTP (return-to-player) configuration by redesigning the loading flow to externalize metadata and adding ability to override the RTP display on game launch.
- Released new game title, All Aboard Panda Pennies, by cloning an existing game and extending it with responsive orientation support, full-screen background handling, and character group animations for compatible with its unique large-reel feature.
- Delivered 10+ staging site launches and 70+ game deployments by managing site-specific configurations across multiple jurisdictions.
- Shipped 15+ game re-releases, resolving 85+ issues to ensure production stability.

Projects

Kotoba Tag!

kotoba-tag.com

- Implemented answer validation with 95% accuracy by training a language model on 4M+ text pairs.
- Deployed a React application with an XState machine that manages game states and core gameplay.
- Created a FastAPI endpoint to offload model inference and send predictions to game client.
- Implemented syllable matching logic in TypeScript according to game rules validated by Vitest unit tests.
- Detailed rules and project objectives in comprehensive README as a reference and design document.

Personal Website

npach.github.io

- Created website hosted on GitHub with a responsive design using HTML, TailwindCSS, @shadcn/ui.
- Designed cursor-trailing animation in the hero section using React hooks.
- Implemented contact form with Zod validation and the EmailJS API.

Image Analysis: Skin Detection

Spring 2024

- Researched color analysis in skin detection, color spaces and techniques in dynamic thresholding.
- Enhanced existing Python algorithm by refactoring for improved readability, ease of use and relevance.
- Implemented asynchronous image processing for decreased runtime.
- Created Metrics class to calculate algorithm performance, yielding a skin detection accuracy of 98%.

Education

University of Massachusetts Boston

May 2024

B.Sc in Computer Science

Supervised Machine Learning: Regression and Classification

January 2024

Coursera (DeepLearning.AI, Stanford University)